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Efficient and Quantum-safe Internet Key Exchange for Satellite Communications

D. De Zuane[1] P. Santini [2] M. Baldi[2] G. Anchelergues[3]
D. Romano[3] A. Cammarano[3] J. J. Grosso[3]

[1]IMT School for Advanced Studies, Lucca, Italy
[2]Università Politecnica delle Marche, Ancona, Italy
[3]OSMIUM, Turin, Italy

davide.dezuane@imtlucca.it

Outline

01 **Problem Statement**
Post quantum cryptography, internet protocols & satellites

02 **Background**
IKEv2 protocol, post-quantum cryptographic primitives

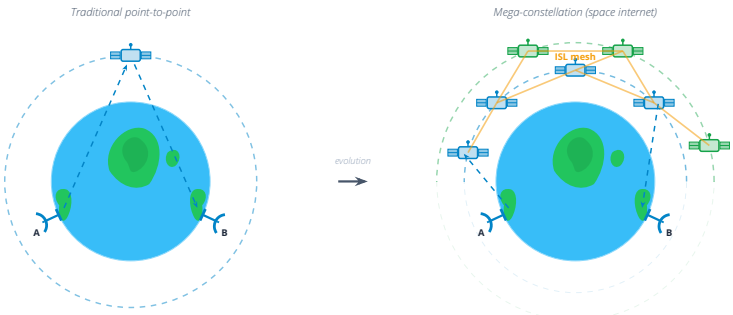
03 **Proposed Solution**
Lightweight variants of IKEv2

04 **Performance Evaluation**
Latency, bandwidth overhead, and computational cost

05 **Conclusions**
Pros & cons, trade-offs and future directions

Evolution of Satellite Communications

Satellite networks from *point-to-point* links → *space-based analogue of Internet*.



⚠ What makes satellite networks different from the Internet

- **High and variable latency**: every extra protocol round-trip multiplies these costs.
- **Resource-limited hardware**: optimised for mass and power, not for performance.

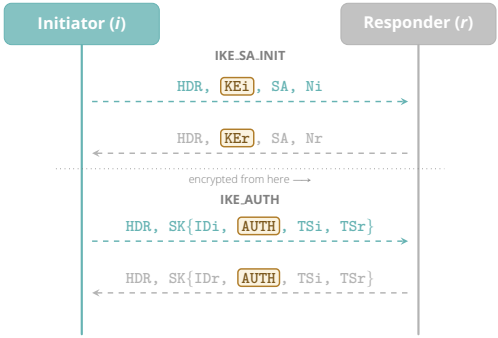
Automated Key Exchange and IKEv2

Cryptographic key distribution:

- Manual pre-sharing does **not scale**.
- Already addressed in early versions of the Internet.
- Starting point:

Internet Key Exchange

- Defined in RFC 7296 [1]
- Authenticated Key Exchange
- Confidentiality provided by **KEX** in INIT
- Authentication provided by **DSS** in AUTH
- Multi round trip protocol



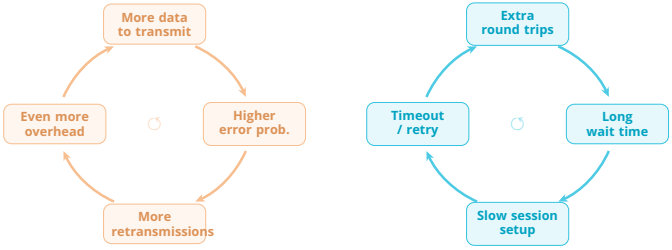
? Question

So we already find the solution for our problem?

IKEv2 over Satellite Networks: Main Problem

Satellite links as a medium, introduce two key challenges:

- **Error loop**: higher error rate ⇒ larger messages fail more often ⇒ retransmissions ⇒ more traffic
- **Latency loop:** multi-RT handshake × ~600 ms RTT ⇒ setup time grows with every exchange



⇒ IKEv2 **impractical** over satellite

⇒ **even worse** when considering post-quantum cryptography

Solution

A **protocol redesign** is therefore necessary.

IKEv2 over Satellite Networks: Challenges

Latency - Protocol Complexity

- Many optional extensions
- Dynamic algorithm negotiation overhead
- Reduce round trip

Minimal IKE [2] only strictly necessary features

Bandwidth - Quantum Payload Growth

- Replace classical asymmetric primitives
- A simple drop-in replacement is **not possible**
- Computational cost comparable but payload size explosion

Accommodate the primitive within the protocol.

KEX Algorithm	pk	ct	
X25519	32 B	32 B	T
ML-KEM-512	800 B	768 B	PQ
ML-KEM-768	1 184 B	1 088 B	PQ
McEliece-348864	261 KB	128 B	PQ

DSS Algorithm	pk	sig	
ECDSA P-256	64 B	64 B	T
RSA-3072	384 B	384 B	T
ML-DSA-44	1 312 B	2 420 B	PQ
ML-DSA-65	1 952 B	3 293 B	PQ

- T stands for Traditional cryptography
- PQ stands for Post Quantum cryptography

Technical baseline and redesign of IKEv2

To make the protocol suitable for satellite communications we target :

- Reduce the number of RT as much as possible
- Make both phases of the protocol (`INIT` and `AUTH`) quantum-safe
- Use Hybrid PQ/T Cryptography for the transition period

Technical Baselines

- `TB1` — IKEv2 with RFC 9370 [3]
  Hybrid KEM via INTERMEDIATE, classical DSS
- `TB2` — IKEv2 with PSK and ML-KEM [4]
  Full PQ in 2 RT, but PSK does not scale

Baselines represent the best achievable within current standards — used as reference.

⇒ we design **three lightweight variants** (LW1, LW2, LW3) that progressively push the optimisation further.

LW1 - Full Capability Mode

Design Choices

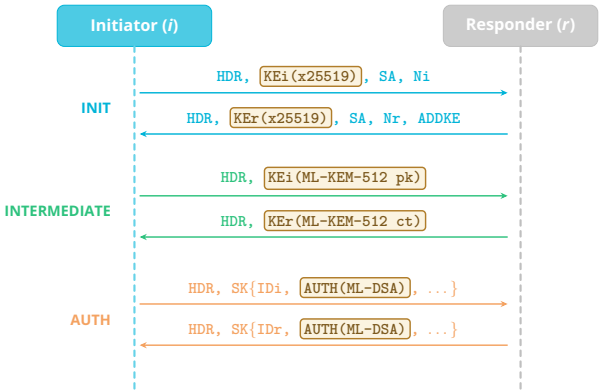
- Hybrid KEX via RFC 9370
- Classical DSS → PQ DSS (ML-DSA-44)
- DSS PK material pre-shared **OOB**

Pro

- Full **hybrid KEX**
- Full RFC 7296 compliance
- **Partial** hybridization

Cons

- Large AUTH messages signature: 2420 B
- No Hybrid Authentication
- Extra Round-Trip



LW2 - Minimal Overhead Mode

Design Choices

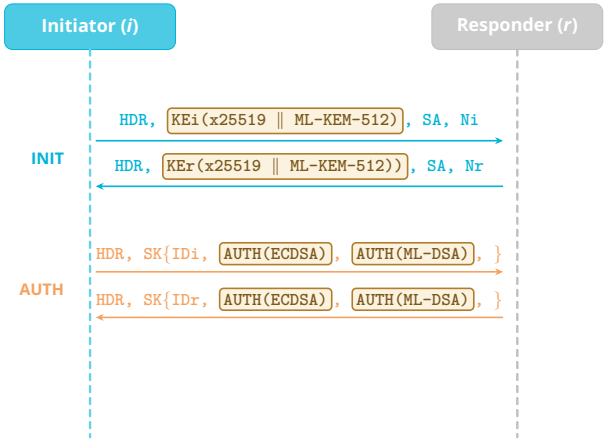
- Hybrid KEX via **combiner**: C‖PQ
- Hybrid Auth with two separate AUTH payloads
- DSS PK material pre-shared **OOB**

Pro

- **Full** hybridization
- Only **2 Round Trip** to complete

Cons

- Breaks RFC 7296 compliance
- MTU constraint caps security



LW3 - KEM-based Minimal Exchange

Design Choices

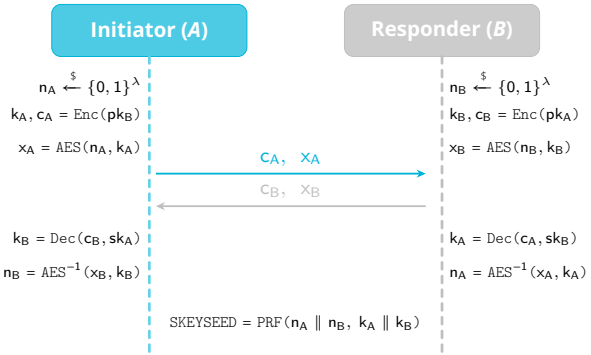
- Opinionated approach
- KEM-only exchange
- KEM PK material pre-shared **OOB**

Pro

- Only **2 messages**
- Smallest footprint

Cons

- No **Perfect Forward Secrecy**
- No cryptographic agility
- Breaks RFC 7296 compliance



Summary

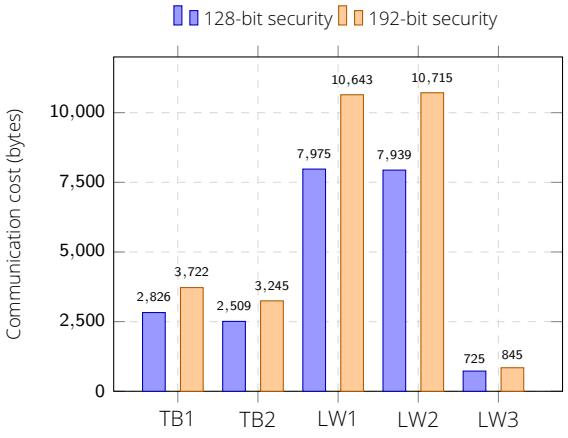
Table: Summary of IKEv2 variant configurations for satellite communications

Variant	Exchanges	Post Quantum	Hybridization	Authentication Method	ROHC	Bandwidth Impact	PFS	IKE Compliance
TB1	`INIT + INTE + AUTH`	◐	RFC 9370 [3]	Traditional DSS Raw Public Keys	✗	High	✓	✓
TB2	`INIT + AUTH`	●	✗	Symmetric PSK	✗	Minimum	✓	✓
LW1	`INIT + INTE + AUTH`	●	RFC 9370 [3]	PQ DSS X.509 Certificate	✓	Medium	✓	✓
LW2	`INIT + AUTH`	●	Crockett et al. [5]	PQ/T DSS Raw Public Keys	✓	Low	✓	✗
LW3	`INIT + AUTH`	●	✗	Implicit via out-of-band	✓	Very low	✗	✗

● Post-quantum in both INIT and AUTH ◐ Post-quantum in INIT only ○ No post-quantum

Communication cost

Total bytes exchanged to complete the handshake, measured across all protocol messages at NIST levels 1 and 3.



LW3 - ultra compact

Only ~725 B at L1. Two McEliece ciphertexts (128 B each) plus encrypted nonces.

11× smaller than LW1/LW2.

LW1 / LW2 — DSS overhead

Cost driven by material transmitted in-band

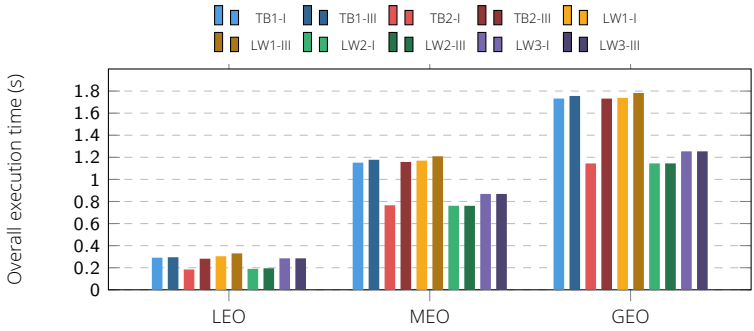
- `ML-DSA` signatures (2 420 B at L1)
- and `ML-KEM` public keys (800 B).

TB1 / TB2 — no PQ DSS

Lower cost because authentication relies on classical ECDSA (64 B) or PSK-HMAC - no large PQ signature payload.

Experimental Assestment in Satellite scenario

Average time to complete the full handshake (30 runs) across LEO, MEO and GEO topologies.



- **LEO:** RT cost does not dominate — all variants perform comparably.
- **MEO/GEO:** saving one RT outweighs larger payload.
- **LW1** pays ~40% extra time vs **LW2** on GEO due to the additional INTERMEDIATE exchange.

Conclusions

- **LW2** matches **TB2** execution time while adding full hybrid KEM + DSS at the cost of **RFC 7296 compliance**.
- **LW3** achieves the **minimum communication cost**, but has some open problems.
- Heterogeneous variants, each fits a *different deployment scenario*

Future works

- Revise LW3 to provide PFS
- `IND-CPA` security for the KEM (an example lean-BIKE [6])
- Extend evaluation to other PQ primitives as standards evolve.

Acknowledgements

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Thank You!

Questions?

davide.dezuane@imtlucca.it